FemboyEngine2D API Reference Manual

C++26 data-oriented engine - complete public surface, contracts and a build-from-zero walkthrough

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1. Scope, conformance and engine rules

This manual specifies the public surface of FemboyEngine2D: every header under Source/Engine and the editor infrastructure that a host links against. It is written against the release-candidate tree, so a symbol described here exists in the headers that ship with that tree, and the appendix index is generated from those headers on every run.

The engine targets C++26 with GCC 16.2.0 or newer and Clang 23.1.4 or newer; the Windows release candidate is built with clang-cl against the MSVC standard library. Four rules are load bearing and every public API is designed around them:

- Simulation state is plain, trivially copyable data. No virtual calls, no owning pointers and no std::string belong in a component, snapshot or network payload.
- Hot paths do not allocate. Frame scratch memory comes from FrameArena, logs go through fixed char buffers, and callbacks are function pointers plus void*, never std::function.
- Determinism is a property of the data, not of the scheduler. Every subsystem exposes a checksummed image, and NetRollbackSession can only replay what it can reproduce byte for byte.
- No RTTI in engine code: dynamic_cast and typeid are rejected by the style gate. Cast<> walks stable type ids instead (see section 3).

The style gate (Source/Tools/Dev/CheckStyle.py) enforces the naming and structure rules: PascalCase file stems, Allman braces, braced control bodies, In-prefixed parameters, Scoped-prefixed locals, no trailing line comments and no Cyrillic in engine sources.

2. Building, targets and compiler flags

One CMake tree produces three kinds of artefact: the engine static library (FemboyEngine2D::Engine), the editor library and application (FemboyEngine2D::Editor), and the offline tools. A shipped game links the engine only; Dear ImGui, ImGuizmo, ImPlot and the TOML writer never enter a shipping target.

CMake option	Default	Effect
FBY_BUILD_EDITOR	ON	Builds fby_editor, the editor app and the panel plugins.
FBY_BUILD_TESTS	OFF	Builds the ctest suite under Source/Tests.
FBY_BUILD_TOOLS	ON	FempakCooker, LocTool, SceneTool, ShaderCompiler, AotCompiler.
FBY_BUILD_PLATFORM_PROVIDERS	ON	Steam and Epic Games Store provider plugins.
FBY_RHI_BACKEND	auto	auto, vulkan, metal or null. auto selects per platform.

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo \
      -DFBY_BUILD_TESTS=ON -DFBY_BUILD_EDITOR=ON
```

```
cmake --build build --target fby_editor
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

The compiler wrapper (cmake/FbyCompiler.cmake) owns the C++26 flag, the warning set and the sanitizer switches. Do not add flags per target: a target that needs a different dialect is a target that should not exist.

3. Core value types, FORCEINLINE and the cast family

Definitions.hpp is the contract header of the engine. It supplies the export macros, the assertion macros (FBY_ASSERT, FBY_VERIFY, FBY_UNREACHABLE), the stable TypeId machinery, the cast family and the component registration macros. Every other header includes it, directly or transitively.

```
#if defined(_MSC_VER)
#define FORCEINLINE __forceinline
#elif defined(__GNUC__) || defined(__clang__)
#define FORCEINLINE inline __attribute__((always_inline))
#else
#define FORCEINLINE inline
#endif
```

FORCEINLINE is the only inlining spelling in public headers. It is used for accessors, hot arithmetic and template casts; it never replaces a measured optimisation. The three families are:

Symbol	Contract
FemCast<To>(value)	Unchecked static_cast with reference collapsing. Use when the concrete type is proven by construction (factory output, storage tag). No runtime cost.
Cast<To>(object)	Checked cast. Upcast is a compile-time static_cast; downcast compares the object's GetTypeId() against StableTypeId<To>() and its registered parent chain; unrelated types fail to compile. Returns nullptr on a mismatch, never throws, never uses RTTI.
CastComponent<To>(registry, entity)	ECS spelling of Cast<>: the type system proves the component type at compile time and the registry validates the entity. Returns nullptr when the entity lacks the component.

```
// Downcast through the stable type table, not through RTTI.
Fby::TypeObject* base = factory.Create("Sprite");
if (Fby::Sprite* sprite = Fby::Cast<Fby::Sprite>(base))
{
    sprite->Animate(dt);
}
```

```
// Registration binds Sprite to TypeObject and derives its stable id.
FBY_REGISTER_TYPE(Fby::Sprite, Fby::TypeObject);
```

StableTypeId<T>() is a consteval FNV-1a of the type's registered name (or the compiler signature when no name is pinned). A type that changes its registered name changes its id, which is why component names are part of the serialized schema and why Snapshot v5 stores a LayoutHash next to the name.

4. FrameArena: the only frame allocator

FrameArena is a bump pointer over caller-owned storage. It is reset at the start of a frame and never frees individual allocations. Systems that need transient arrays take it from SystemContext::frameArena; a system that calls operator new in its frame path is a defect, not a style preference.

```
Fby::FrameArena arena(storage);          // caller owns storage
arena.Reset();                           // once per frame
auto* items = arena.AllocateArray<MeshRenderItem>(count);
if (items == nullptr)
{
    // Arena exhausted: degrade, never allocate on the heap.
    return;
}
const std::size_t used = arena.Used();
const std::size_t capacity = arena.Capacity();
```

- Allocate() returns nullptr instead of throwing when the arena is exhausted; callers must handle the failure.
- AllocateArray<T>() static_asserts std::is_trivially_copyable_v<T>: the arena stores POD payloads only.
- Over-aligned types must pass their alignment explicitly; the default is alignof(std::max_align_t).

5. JobSystem and per-thread diagnostics

JobSystem owns a pool of std::jthread workers. Tasks are function pointers plus a context pointer plus a half-open range, so submitting a task never allocates. ParallelFor splits a range into chunks, lets every participant pull the next chunk from an atomic counter, and always runs at least one chunk on the caller, which keeps single-core machines correct and worker re-entry safe.

```
jobs.ParallelFor(entityCount, 4096, [&](std::uint32_t begin, std::uint32_t end)
{
    for (std::uint32_t i = begin; i < end; ++i)
    {
        Integrate(entities[i], dt);
    }
});
jobs.WaitIdle();
```

```
// Diagnostics: one row per thread, slot 0 is the main thread.
std::array<Fby::JobSystem::ThreadLoad, Fby::JobSystem::kMaxThreadSlots> loads{};
const std::uint32_t slots = jobs.SampleThreadLoad(loads);
```

ThreadLoad carries busyNanoseconds (wall time inside engine work), taskCount and a 16-byte name. The counters are one cache line per thread, so two workers finishing at the same time do not contend. SceneDiagnosticsPanel turns two samples into a per-thread occupancy percentage.

ThreadSlotCount() is ThreadCount() + 1 and is clamped by the fixed MCounters array; a device with more than 64 workers reports the first 64 slots.

6. ECS, scheduling and snapshots

The registry is FCS. Systems are registered on a SystemGraph in one of five phases (PreUpdate, FixedUpdate, Update, PostUpdate, Render). SystemContext is the only parameter a system receives; it carries the registry, JobSystem, frame arena, renderer, window, VFS, AssetDatabase, audio, physics, joints, cloth, fluid, camera stack, accessibility kernel and font atlases.

```
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem, nullptr);
schedule.Add(Fby::Ecs::Phase::PostUpdate, "platform", &GogAchievementSystem,
&gog);

Fby::Ecs::SystemContext context{registry, jobs};
context.physics = &physics;
context.window = &window;
schedule.Run(Fby::Ecs::Phase::FixedUpdate, context);
```

Snapshot v5 is a canonical, checksummed byte image of the registry. Components are written by hand-written codecs when they contain runtime handles (b2BodyId, b2ShapeId); the snapshot never serializes those handles, and the restore path recreates bodies from the component fields. The schema stores the Mirror name and LayoutHash of every component, so a field change is rejected instead of silently misread.

API	Contract
GetOrAdd<T>(registry, entity)	Adds the component when missing and returns a stable reference.
ecs::SnapshotWriter / SnapshotReader	Writes and validates the canonical image; checksum is FNV-1a over the payload.
Fby::Net::NetRollbackSession	Consumes snapshot v5 and replays ticks on late input (section 10).

7. RHI, render items and frame statistics

The RHI is a hand-written Vulkan/Metal layer. Frames are recorded into persistent ring buffers; sprite items are 48-byte PODs; meshes are drawn through one indirect call per material/layer; linear-blend skinning is one compute dispatch. FrameStats exposes what the pass submitted, which is what SceneDiagnosticsPanel plots.

FrameStats field	Meaning
items / batches / drawCalls	Sprite items, batches and submitted draws.
meshItems / meshIndirectCalls	MeshRenderItem count and indirect draws.
skinDispatches / skinRanges	Skinning compute dispatches and ranges.
uploadedBytes	Bytes uploaded this frame.

The editor's Viewport renders into an offscreen target, not into the swapchain. EnsureViewportTarget() sizes that target in physical pixels; a target sized in points is the usual cause of a blurry Viewport on a HiDPI monitor.

8. Materials: cooked graphs, the pipeline cache and the shipped path

Material authoring and material compilation live on opposite sides of the engine boundary. `Render::MaterialGraph` is the authoring DAG: fixed capacities (128 nodes, 384 pins, 64 parameters), trivially copyable, and hashed without the editor positions. `Render::MaterialGraphCompiler` turns a validated graph into GLSL 450 text. `fby_matc` runs that compiler and glslang at build time and writes a `.femmat` payload, so a shipped game never links glslang and never parses GLSL. The full rationale is in [ADR 0002](../../adr/0002-materials-in-engine.md).

Piece	Header	Contract
`Render::MaterialGraph`	`Engine/RHI/MaterialGraph.hpp`	Authoring DAG; `Hash()` is the pipeline identity and excludes UI positions
`MaterialGraphCompiler`	`Engine/RHI/MaterialGraphCompiler.hpp`	DAG to GLSL 450 and the shader contract flags
`EncodeMaterialGraph` / `DecodeMaterialGraph`	`Engine/RHI/MaterialAsset.hpp`	`.fgraph` wire format: explicit little-endian fields, no struct memcpy
`EncodeCookedMaterial` / `DecodeCookedMaterial`	`Engine/RHI/MaterialAsset.hpp`	`.femmat` payload: 32-byte header, vertex and fragment SPIR-V, zero-copy decoding
`Render::MaterialPipelineCache`	`Engine/RHI/MaterialPipelineCache.hpp`	128 pipelines keyed by graph hash, reference counted, LRU eviction, no heap use
`Rhi::RHI::SupportsRuntimeShaderPipelines`	`Engine/RHI/RHI.hpp`	Capability query: Vulkan returns true, Null and Metal return false

```
# Author the graph once, cook it in the build step
build/bin/fby_matc assets/materials/tinted.fgraph -o
assets/materials/tinted.femmat
```

```
// `InBytes` is the .femmat payload from the VFS or a package entry.
Render::CookedMaterialView ScopedMaterial;
if (!Render::DecodeCookedMaterial(InBytes, &ScopedMaterial))
{
    return false;
}

// One cache per device: acquire at load time, release at shutdown.
const Rhi::ShaderPipelineHandle ScopedPipeline = InMaterials.Acquire(InRhi,
ScopedMaterial);

Rhi::MeshRenderItem ScopedItem;
ScopedItem.materialPipeline =
    ScopedPipeline.IsValid() ? ScopedPipeline.id :
Rhi::kDefaultMaterialPipeline;
```

- The graph hash is the pipeline identity. Editing a graph produces a new hash, so the old pipeline becomes an unreferenced cache entry instead of being mutated in place.
- The contract mask travels with the payload: generated vertex stage, fragment view projection, material push constants, world and vertex position varyings. A live entry whose hash arrives with another contract is refused, because that is a different shader.
- `Acquire()` adds a reference and `Release()` drops one. A pipeline at zero references stays resident, so a material acquired once per frame never recreates a pipeline; it is destroyed by LRU eviction, `ReleaseAll()` or `Shutdown()`.
- The cache never allocates and never formats text on the frame path; `LastError()` returns the fixed error buffer.
- Backends without runtime shader pipelines keep the engine shaders. Query `SupportsRuntimeShaderPipelines()` instead of probing for an invalid handle.

Payload	Fields	Validated on load
`.fgraph`	magic `FBYG`, version, node/pin/param counts, nodes, pins, params	magic, version, counts against capacity, exact size, node kinds, pin and param indices
`.femmat`	magic `FBYM`, version, contract mask, graph hash, vertex and fragment byte lengths	magic, version, size against the lengths, both SPIR-V modules, contract consistency

The editor keeps live editing on top of the same cache: `Fby::Editor::MaterialGraphRuntime` recompiles a graph when its hash changes, reports per-tab errors and delegates pipeline ownership to `MaterialPipelineCache`. `MaterialGraphRuntime::SetDomain` switches the live compile between the mesh and the sprite contract, and the panel's `SaveGraph`/`LoadGraph` write and read the

`.fgraph` payload the cooker consumes. A shipped build skips glslang entirely and only calls `DecodeCookedMaterial` and `Acquire`.

Sprites take materials through the third artefact: `MaterialCompileOptions::domain` selects the sprite domain, where the generated fragment stage matches the `quad.vert` varyings (`vUv`, `vColor`, `vLayer`, and `vWorld` when a node needs it) and samples the item's atlas slice through `vLayer`. `Time` is available: the sprite push block carries a material clock at offset 80, so the generated stage reads `pc.materialParams.x` exactly like a mesh shader. `fby_matc --sprite` cooks that variant. `DrawOrder::MaterialBucket` places the sprite in a bucket, `ChangeMaskRenderExtract::SetMaterialPipelines` maps buckets to cached pipeline ids, and the extractor emits one `DrawBatch` per bucket with `DrawBatch::pipeline` already resolved. The 64-bit render key carries the bucket (16 layer, 8 material, 32 depth, 8 sequence), so a material batch is never split by depth. Nodes that need the mesh view-projection (`Fresnel`, `VertexPosition`) are refused in the sprite domain with a readable reason.

9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem

PhysicsSystem owns one b2WorldId and exposes it to extension modules as a packed uint32, so box2d.h never leaks into a public header. Bodies are mirrored from RigidBody2D/Collider2D components into Box2D, stepped at the fixed rate, and read back before interpolation. PhysicsJointSystem builds a fixed-capacity joint graph: chain, rope and spring connections with break limits, damage accumulation, gameplay cuts, repair, in-place rebuild and ties. ClothSystem integrates a Verlet grid and projects it against the same Box2D colliders.

SdfCollisionSystem contract:

Member	Contract
SetField(const SdfField&)	Installs a non-owning mdspan over a row-major grid. Negative values are solid; the field's memory is owned by the renderer/terrain, not by the system.
Step(registry, physics, fixedDt)	Samples the grid at body corners and centre, converts the gradient to a normal, applies a mass-scaled push impulse, reflects inward velocity and writes the corrected state back. Returns the number of corrected bodies.
Update(SystemContext&)	ECS convenience wrapper around Step(); a headless call without physics is a no-op.
Contacts()	Fixed array of SdfContact {entity, point, normal, penetration, impulse, sequence}.
SdfCollisionDesc	pushOut, restitution, maxImpulse, subSamples, dynamicOnly.

```
Fby::SdfCollisionDesc desc;
desc.pushOut = 0.65f;
desc.restitution = 0.05f;
desc.subSamples = 2;
```

```
sdf.Initialize(desc);
sdf.SetField(terrain.Field());
sdf.Register(graph);                    // Phase::FixedUpdate

// Viewport overlay: red vectors whose length is the penetration.
debug.Build(context, Fby::Editor::PhysicsDebugSdf, &sdf);
```

SdfCollisionSystem is not part of SystemContext. A host that wants SDF physics registers the system explicitly through Register(graph) and keeps the instance alive; the Viewport debug pass takes the same pointer.

10. NetRollbackSession: deterministic rollback

A session binds snapshot v5, CommandLog, Pcg32 and the gameplay components (tags, attributes, effects and skills) into one fixed-capacity rollback loop. Everything after Initialize() is allocation-free: one snapshot ring, one input ring per player, a fixed entity map and fixed frame metadata. The simulation callback is a function pointer plus void*.

Method	Contract
Initialize(registry, schema, config, arena)	Allocation-free setup with a caller-owned snapshot arena.
Initialize(registry, schema, config)	Convenience overload; allocates one arena (the only allocation the session makes).
SetSimulation(fn, user) / SetCommandLog(log)	Binds the pure simulation function and the structural command log.
Start(rng)	Captures the base tick and stores the session RNG. Required before Step().
Step(localInput)	Simulates exactly one tick with prediction for missing remote inputs and captures the snapshot.
SubmitRemoteInput(player, frame)	Stores a late or current input; a frame for an already simulated tick marks a pending rollback.
ReplayPending(report)	Restores the snapshot before the earliest corrected tick, restores RNG and CommandLog cursor, re-simulates, captures new snapshots and reports changed ticks.
CompareChecksums(tick, snap, state, out)	Compares a remote report with the stored frame; the two checksums isolate a byte-level desync from a gameplay-state desync.

```
Fby::Net::NetRollbackSession::Config cfg;
cfg.playerCount = 2;
cfg.localPlayer = 0;
cfg.rollbackTicks = 64;
session.Initialize(registry, schema, cfg, arena);
```

```
session.SetSimulation(&SimulateTick, &game);
session.Start(pcg);

while (running)
{
    session.SubmitRemoteInput(remote, received);
    session.Step(localInput);
    if (session.RollbackPending())
    {
        Fby::Net::RollbackReport report;
        session.ReplayPending(&report);
        if (report.changedTicks > 0)
        {
            LogBoy().Log(Fby::Editor::LogBoySeverity::Desync, "net",
                         "rollback {}..{} changed {}", report.fromTick,
report.toTick,
                         report.changedTicks);
        }
    }
}
```

- NetInputFrame is 24 bytes of integer axes (int16 fixed point). Floats are not allowed: a float bit pattern is not guaranteed to survive a different compiler.
- Result enumerates every failure mode (NotInitialized, BufferFull, RollbackTooOld, ChecksumMismatch, ...); ResultName() is the log spelling.
- A session is non-copyable and owns its arena when the convenience overload is used.

11. Audio: mixer, DSP graph, limiter

AudioSystem owns the miniaudio device and a fixed mixer. DspGraph routes nodes into buses, supports ducking and reverb, and receives commands through a lock-free ring so the game thread never blocks the audio callback. Limiter.hpp is the final true-peak stage.

- DeafMode mutes the graph at the Master root and turns sound triggers into subtitle lines; it is a policy of the audio system, not a post-effect.
- Ogg/Opus is decoded off the audio thread; the mixer only sees cooked .fpcm buffers.

12. PinkUI and WidgetCanvasSystem

PinkUI is the in-game UI: a fixed-capacity canvas, mdspan grid layout, anchor resolution, hit testing and one-batch packing. WidgetCanvasSystem turns a WidgetCanvas into FrameOutput (draw items) without allocating; font glyphs come from a .fbyfont atlas mapped through the VFS, so the shipping runtime carries no TTF rasterizer. Glyphs are atlas-only: there is no embedded 8x8 bitmap fallback, and a text frame without a mounted font array is refused instead of drawn with the wrong layer.

Member	Contract
WidgetCanvasSystem::MountFonts(RHI&)	Mandatory cold-start stage on the main thread. Builds the GPU texture array from the

	VFS .fbyfont set; a failure is a PANIC log plus FBY_ASSERT_MSG, never a fallback.
BuildFrame(context, canvas, output)	Lays out and culls one canvas, returning BuildStatus and filling FrameOutput. A text canvas without a current mounted atlas returns FontsNotMounted and writes nothing. Allocation-free; safe to call from Phase::Render.
Update(context, canvases, output)	Builds every active canvas into one output; the host submits the result as one batch. A font-contract violation stops the pass and sets fontContractViolated.
WidgetCanvas::SetDisplayScale	Density from CameraStackSystem::DisplayScale; font sizes are rounded to whole physical pixels.
OledShiftState	Integer 2 px orbit for OLED safety; ApplyOledShift() binds it to a canvas pool.

WidgetCanvasSystem is also the accessibility surface: BlindMode reads widget labels for the narrative log, DeafMode receives subtitle widgets, and colour-vision matrices plus FlashGuard are applied to the canvas colours before packing.

13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)

Accessibility is a frame policy, not an overlay. BlindMode skips the game pass, indirect draws and the skinning dispatch, then publishes a bounded narrative log. DeafMode mutes the Master bus and routes sound triggers into SubtitleLines. FlashGuard compresses sudden luminance changes. Colour-vision correction uses the published Machado 2009 matrices, and SrgbRelativeLuminance()/ContrastRatio() implement WCAG 2.2 for authoring-time validation.

Item	Contract
NarrativeEvent	128-slot SPSC ring of 160-byte text payloads; the simulation thread pushes, the speech thread pops.
ColorVisionMatrix(deficiency)	Returns the 3x3 matrix for protanopia, deuteranopia or tritanopia.
ApplyColorVision(color, matrix, strength)	Applies the matrix in linear light to an sRGB Color32.
FlashGuard::Evaluate(luminance, dt)	Returns the gain for the current frame; a large delta is compressed to the threshold.
EditorSettings::ide*	The same policies applied to the editor window itself: Machado correction, WCAG high-contrast, FlashGuard, keyboard navigation and LogBoy as the narration sink.

14. Asset pipeline, VFS and localization

Assets are addressed by relative path inside the VFS. A .fempak archive is mounted with a priority; a directory mount can shadow it, which is how the editor writes a loose file before the cooker folds it back in. Read() returns a span into an mmap that stays valid until the mount is dropped or the file is hot-reloaded.

Format	Owner	Purpose
.fempak	FempakCooker	Immutable shipped archive with an index and optional encryption.
.fbyfont	FontAtlas (PinkUI)	Baked glyph atlas: [header] [glyph records][pixels]; no runtime TTF rasterizer.
.femeta	AssetDatabase	Typed metadata, import settings and stable AssetId.
.fbin	LocTool	Compiled localization table; the editor UI itself is English-only.

The editor UI is not localizable by design: menus, panel titles, errors and the inspector are English-only. Localization applies to game content and documentation, where a .fbin table and the EN/RU manual cover the shipped languages.

15. Platform services: Steam, Epic Games Store, GOG

PlatformService is the engine-side bridge; PlatformAPI.h is a flat pure-C ABI that a store SDK plugin implements. The provider is loaded dynamically, resolves only documented entry points and converts every store-specific type at the boundary, so a game never includes steam_api.h, EOSSDK or Galaxy headers.

Provider	Status in this tree	Notes
SteamProvider.cpp	Implemented	Steamworks flat API through a function table.
EgsProvider.cpp	Implemented	Epic Online Services provider through the same C ABI.
GogProvider.cpp	Implemented (shim)	Loads galaxy64.dll / libgalaxy64.so / libgalaxy64.dylib at runtime and resolves the flat fby_gog_* shim ABI; the game ships the shim built against its SDK revision. Galaxy is a C++ interface library, so a shim is mandatory. GogAchievementSystem and GogStorageSystem write the

		binary PlatformCommandLog in Phase::PostUpdate.
fby_fontc	Implemented	Offline TTF -> .fbyfont cooker (stb_truetype, RGBA8 atlas, verified header).

```
// Pure C ABI: the provider is a shared library, not a C++ dependency.
typedef struct FbyPlatformApi {
    uint32_t version;
    void* user;
    bool (*init)(void* user);
    void (*shutdown)(void* user);
    void (*unlockAchievement)(void* user, const char* id);
    bool (*writeCloudFile)(void* user, const char* name, const uint8_t* data,
uint32_t size);
} FbyPlatformApi;
```

16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics

The dashboard is the only editor surface that exists before a project is open. It installs the editor font set before the first frame, so the first window is already Inter at the physical DPI. A new project starts in Projects/ under the engine root, resolved by DefaultProjectsRoot() - never in the process working directory and never in the install directory.

Component	Contract
EditorFonts	Loads Inter and JetBrains Mono from the VFS mmap. Subpixel advances (PixelSnapH=false), auto oversampling, no bitmap fallback, dynamic atlas baked at the viewport pixel density. Legacy backends get RasterizerDensity + inverse FontGlobalScale.
ProjectDashboard	Initialize(paths, config, visible, window, vfs, fonts, uiScale): installs typography, resolves the default Projects/ location and creates it.
LogBoySystem / LogBoyPanel	Fixed 4096-record ring of 192-byte text payloads with severity Trace/Info/Warning/Critical/Desync, FNV-1a system hashes and colour+hash filtering.
SceneDiagnosticsPanel	Per-thread occupancy bars and heatmap, FrameArena bytes, MeshRenderItem count, indirect draws, draw calls and exact simulation microseconds with avg/p99/peak.
PhysicsDebugVisualizer	Fixed-buffer line/point geometry for Box2D bodies, ropes, cloth and SDF normals, drawn over the Viewport through a caller-supplied world-to-screen map.

The .fbyfont atlas is a baked PinkUI asset and cannot feed ImGui's vector rasterizer. The editor therefore loads the Inter TTF from the same VFS mount; the atlas header is still validated on startup, so a corrupt game-UI atlas is reported by the editor, not by the player.

17. Step-by-step: a game from an empty folder

This walkthrough produces a playable project with a sprite, physics, keyboard input, a PinkUI HUD and a deterministic rollback session. Every step maps to a real header from the index.

- 1. Launch the editor and create a project with New Project. The dashboard creates <engineRoot>/Projects/<name> with assets/{textures,audio,scenes}, configs/ and project.femproj.
- 2. Import a sprite into assets/textures. The watcher imports it, writes a .femeta and cooks a .ftex texture array layer.
- 3. Create assets/scenes/main.toml with one SpriteRenderer, one Transform2D and one RigidBody2D + Collider2D.
- 4. Register the systems: input in PreUpdate, physics in FixedUpdate, movement in Update, PinkUI in Render.
- 5. Give the camera a CameraStackSystem and call SetDisplayScale() from SDL_GetWindowDisplayScale so UI glyphs are crisp on HiDPI.
- 6. Add an AccessibilityKernel from project settings; BlindMode and DeafMode are read from project.femproj, not hard-coded.
- 7. Create a NetRollbackSession per match, bind a pure SimulateTick callback and feed NetInputFrame values.
- 8. Log through LogBoy: severity + system hash + fixed-format message. Open the LogBoy panel and filter by Desync while testing netcode.
- 9. Enable Physics Debug Draw in the Viewport while tuning colliders and cloth.
- 10. Cook with FempakCooker, run the suite with ctest and the style gate, then ship the engine-only target.

```
// 4-7 in code: systems, accessibility and rollback for one match.
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::PreUpdate, "input", &InputSystem);
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem);
schedule.Add(Fby::Ecs::Phase::Render, "ui", &UiSystem, &ui);

Fby::AccessibilityKernel kernel;
kernel.Configure(settings);              // blind/deaf/colour/flash

Fby::Net::NetRollbackSession session;
if (session.Initialize(registry, schema, cfg, arena) !=
Fby::Net::NetRollbackSession::Result::Ok)
{
    LogBoy().Log(Fby::Editor::LogBoySeverity::Critical, "net", "rollback init
failed");
```

```
}
```

18. Testing, style gate and release checklist

The suite uses no test framework: every test is an executable with a main() that returns non-zero on failure, so it runs on a headless CI agent, a console devkit and a developer workstation with the same command. The full suite passes at Release.

```
ctest --test-dir build --output-on-failure          # the full suite
python Source/Tools/Dev/CheckStyle.py               # style gate
python Documentation/Tools/GenerateAPIReference.py  # this manual
python Documentation/Tools/GenerateDocumentation.py
python Documentation/Tools/GenerateApplicationGuide.py
```

- A release build enables asserts and runs the whole suite once with FBY_ENABLE_ASSERTS=1, because two latent debug-only compile errors existed in JobSystem at RC.
- The POSIX branch of WriteFileAtomically() and LogLine() must be compiled on Linux/macOS in CI; the Windows-only build never saw them.
- The style gate must report zero failures before a commit is eligible for release.

19. Open work

Open work is tracked in the project issue tracker.

20. Appendix A - generated symbol index

Generated from the headers on every run. Header and line are the authoritative location.

Editor/Infrastructure (119)

Symbol	Kind	Header	Line
AccessibilityEffectRegistry	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	47
AIStateTreeCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	54
AnimationGraphCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	55
AnimationTimelinePanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	56
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	20
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	57
AssetImportService	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	30
AssetImportService	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	58
Callbacks	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	38
Change	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	126
Codec	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	222
Command	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	236

CommandKind	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	33
CommandStack	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	30
CommandStack	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	59
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatch System.hpp	27
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	39
Config	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	88
ContentBrowserPan el	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	60
CookRequest	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	141
CookResult	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	152
EditorCamera	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	101
EditorDesc	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	72
EditorFontRole	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	22
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	84
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/	7

		EditorTypography.hpp	
EditorFontSpec	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	38
EditorFontStatus	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	67
EditorPalette	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	77
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSettings.hpp	20
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	14
EditorSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	61
EditorSymbolColors	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	61
EditorSymbolRole	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	42
EditorSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	110
EditorThemeId	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	31
Entry	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hp p	52
ExternalClientDesc	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	37
FrameGenerationMo de	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	139
FrameGenerationPol icySetting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	132
HudTargetMode	Infrastructure	Source/Editor/	148

		Infrastructure/ ProjectConfig.hpp	
ImFont	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	10
ImGuiIO	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	11
Impl	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	244
InteractionTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	16
LoadedSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	138
MaterialGraphPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	62
MaterialGraphRuntime	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime .hpp	41
MaterialRuntimeStats	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime .hpp	25
MotionComfortSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	171
NetSession	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	32
NetworkDiagnosticsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	63
OitMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	99
OpaqueRenderMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	92
PendingEdit	Infrastructure	Source/Editor/ Infrastructure/	261

		CommandStack.hpp	
PendingSpawn	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	270
PhysicsDebugFlags	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	19
PhysicsDebugFrame	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	47
PhysicsDebugLine	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	32
PhysicsDebugPoint	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	40
PhysicsDebugVisualizer	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	58
PiePlayerInput	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	27
PlaybackIcon	Infrastructure	Source/Editor/ Infrastructure/ EditorWidgets.hpp	12
PlayInEditor	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	44
PluginHost	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	42
Profiler	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	27
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	67
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	286

ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	64
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	350
ProjectDashboard	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	65
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	68
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	308
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	21
ProjectPluginEntry	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	163
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	182
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	22
ProjectSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	66
Range	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	253
ReleaseCookConfig	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	70
ReleaseCooker	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	158
ReleaseCookProgress	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	121
ReleaseCookResult	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	142

ReleaseCookStage	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	28
ResolvedHudSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	35
ResolvedRenderSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	14
Result	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	43
Result	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	47
SdfCollisionSystem	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	13
Services	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	224
SettingDesc	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	28
SettingKind	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	20
SettingScope	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	11
Slot	Infrastructure	Source/Editor/Infrastructure/MaterialGraphRuntime.hpp	118
Source	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	19
SpawnComponent	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	284
SpawnHeader	Infrastructure	Source/Editor/Infrastructure/	277

		CommandStack.hpp	
SpawnTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	190
StableSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	145
Stats	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatch System.hpp	88
Stats	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	55
Stats	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	95
TemporalUpscaleMo de	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	155
TextureArrayAsset	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	23
TextureBinding	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	168
TextureThumbnailCa che	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hp p	17
UpscaleMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	106
UpscalePolicy	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	115
UpscalerBackendSet ting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	122
UserSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	69

UserSettings	Infrastructure	Source/Editor/ Infrastructure/ UserSettings.hpp	17
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	20
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	15
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	18
VirtualTextureSettin gs	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	37
VirtualTextureSettin gs	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	32
VirtualTextureSyste m	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	22
VisualScriptPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	70
WindowManager	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	16

Editor/Panels (90)

Symbol	Kind	Header	Line
Action	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	25
AIStateTreeCanvasP anel	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	55
AIStateTreeCompile d	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	35
AIStateTreeDocume nt	Panels	Source/Editor/ Panels/	24

		AIStateTreeCanvasPanel.hpp	
AnimationGraphCanvasPanel	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	125
AnimationGraphDocument	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	88
AnimationGraphLink	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	74
AnimationGraphLinkKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	37
AnimationGraphNode	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	49
AnimationGraphNodeKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	29
AnimationTimelinePanel	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	34
AssetImportService	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	24
AudioSystem	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	15
AudioSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	11
CanvasModelOps	Panels	Source/Editor/Panels/NodeCanvasPanel.hpp	21
CommandStack	Panels	Source/Editor/	25

		Panels/ ContentBrowserPanel. hpp	
CommandStack	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	30
ComponentRegistry	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel. hpp	18
Config	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	35
ContentBrowserPan el	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	27
DirectionalBlendSpa cePanel	Panels	Source/Editor/ Panels/ DirectionalBlendSpac ePanel.hpp	24
DragPayload	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	60
EditorFonts	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	20
EditorFonts	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	20
EditorSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	91
EditSource	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	58
EditSource	Panels	Source/Editor/ Panels/ AnimationGraphCanv asPanel.hpp	128
Entry	Panels	Source/Editor/ Panels/ ContentBrowserPanel.	43

		hpp	
FileDialog	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	140
ImNodesContext	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	13
ImNodesContext	Panels	Source/Editor/ Panels/ AnimationGraphCanv asPanel.hpp	15
LevelSettingsPanel	Panels	Source/Editor/ Panels/ LevelSettingsPanel.hp p	12
LicensesPanel	Panels	Source/Editor/ Panels/ LicensesPanel.hpp	11
LogBoyPanel	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	394
LogBoyRecord	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	105
LogBoyRing	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	121
LogBoySeverity	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	33
LogBoySystem	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	186
LogicGraphDocume nt	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	23
MaterialGraphPanel	Panels	Source/Editor/ Panels/ MaterialGraphPanel.h pp	20
Mode	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	84

NetSession	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	6
NetworkDiagnosticsPanel	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	15
NodeCanvasPanel	Panels	Source/Editor/ Panels/ NodeCanvasPanel.hpp	54
ParticleFxPanel	Panels	Source/Editor/ Panels/ ParticleFxPanel.hpp	16
PIEAnimFrame	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	110
PieClientViewport	Panels	Source/Editor/ Panels/ PieClientViewport.hpp	15
PIEFrameSlot	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	302
PlaneView	Panels	Source/Editor/ Panels/ DirectionalBlendSpacePanel.hpp	122
ProjectConfigStore	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	31
ProjectConfigStore	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	24
ProjectDashboard	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	22
ProjectSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	33
QueryKind	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	430
SceneDiagnosticsPa	Panels	Source/Editor/	56

nel		Panels/ SceneDiagnosticsPanel.hpp	
SceneDiagnosticsSample	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	15
ScriptDebuggerPanel	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel.hpp	28
ScriptRuntime	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	16
SdfCharacterPanel	Panels	Source/Editor/ Panels/ SdfCharacterPanel.hpp	9
SinkUser	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	88
SkeletonInspectorState	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	19
SpawnContext	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	68
SpriteAnimationInspectorState	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	21
SystemContext	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	24
SystemContext	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	13
SystemContext	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	13

SystemName	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	189
TextPreviewPanel	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	24
TextureThumbnailCache	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	19
TilemapEditorPanel	Panels	Source/Editor/ Panels/ TilemapEditorPanel.hpp	20
TimelinePanelContext	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	22
TransportGroupPanel	Panels	Source/Editor/ Panels/ TransportGroupPanel.hpp	14
UserSettingsPanel	Panels	Source/Editor/ Panels/ UserSettingsPanel.hpp	16
VirtualFileSystem	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	18
VirtualFileSystem	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	13
VirtualFileSystem	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	15
VirtualTexturePanel	Panels	Source/Editor/ Panels/ VirtualTexturePanel.hpp	18
VirtualTextureSystem	Panels	Source/Editor/ Panels/ VirtualTexturePanel.hpp	10
VisualScriptPanel	Panels	Source/Editor/ Panels/	24

		ScriptDebuggerPanel.hpp	
VisualScriptPanel	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	32
VoiceDevicePanel	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	26
VoiceEffectPolicy	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	17
VoicePresetChoice	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	15
VoicePresetRegistry	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	14
VoicePresetSelection	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	16
VoiceReactiveSfxSettings	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	18
VoiceSession	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	12
VoiceSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	13
WidgetCanvas	Panels	Source/Editor/Panels/SettingsPanels.hpp	18
WindowManager	Panels	Source/Editor/Panels/ProjectDashboard.hpp	14

Engine/AI (7)

Symbol	Kind	Header	Line
AIAgentSettings	AI	Source/Engine/AI/AIController.hpp	17
AIControllerComponent	AI	Source/Engine/AI/AIController.hpp	32
AIControllerSystem	AI	Source/Engine/AI/	96

		AIController.hpp	
AIPerceptionComponent	AI	Source/Engine/AI/AIController.hpp	60
AIPerceptionSettings	AI	Source/Engine/AI/AIController.hpp	49
AIPerceptionSystem	AI	Source/Engine/AI/AIController.hpp	68
GridEntry	AI	Source/Engine/AI/AIController.hpp	83

Engine/Assets (64)

Symbol	Kind	Header	Line
AeadAlgorithm	Assets	Source/Engine/Assets/FemCrypto.hpp	18
AeadKey	Assets	Source/Engine/Assets/FemCrypto.hpp	39
AeadNonce	Assets	Source/Engine/Assets/FemCrypto.hpp	46
AeadTag	Assets	Source/Engine/Assets/FemCrypto.hpp	54
ArchiveBuildOptions	Assets	Source/Engine/Assets/FemArchiver.hpp	102
ArchiveCipher	Assets	Source/Engine/Assets/FemArchiver.hpp	29
ArchiveCodec	Assets	Source/Engine/Assets/FemArchiver.hpp	22
ArchiveEntryView	Assets	Source/Engine/Assets/FemArchiver.hpp	118
ArchiveInput	Assets	Source/Engine/Assets/FemArchiver.hpp	72
ArchiveProgress	Assets	Source/Engine/Assets/FemArchiver.hpp	85
ArchiveProgressSta	Assets	Source/Engine/	78

ge		Assets/ FemArchiver.hpp	
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	129
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	148
ArchiveStatus	Assets	Source/Engine/ Assets/ FemArchiver.hpp	36
AssetDatabase	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	36
AssetDependency	Assets	Source/Engine/ Assets/AssetMeta.hpp	83
AssetHandle	Assets	Source/Engine/ Assets/AssetMeta.hpp	22
AssetImportSettings	Assets	Source/Engine/ Assets/AssetMeta.hpp	45
AssetMeta	Assets	Source/Engine/ Assets/AssetMeta.hpp	61
AssetRecord	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	27
AstcBlock	Assets	Source/Engine/ Assets/ TextureCook.hpp	25
AtlasDesc	Assets	Source/Engine/ Assets/ TextureArray.hpp	31
AtlasLayer	Assets	Source/Engine/ Assets/ TextureArray.hpp	21
AtlasPackError	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	57
AtlasSprite	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	26
AtlasSpriteSource	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	18
AtlasUv	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	41
AudioClipHeader	Assets	Source/Engine/ Assets/	87

		AudioClipFormat.hpp	
BlockExtent	Assets	Source/Engine/ Assets/ TextureFormat.hpp	103
CleanRenderExtract	Assets	Source/Engine/ Assets/ ProjectScaffold.hpp	134
CookedAudioView	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	116
CookedTexture	Assets	Source/Engine/ Assets/ TextureCook.hpp	41
CookedTextureView	Assets	Source/Engine/ Assets/ TextureFormat.hpp	203
CookProgressFlags	Assets	Source/Engine/ Assets/ CookProgress.hpp	31
CookProgressRecord	Assets	Source/Engine/ Assets/ CookProgress.hpp	38
CookProgressStage	Assets	Source/Engine/ Assets/ CookProgress.hpp	18
CookTarget	Assets	Source/Engine/ Assets/ TextureCook.hpp	17
FanmHeader	Assets	Source/Engine/ Assets/ AnimationClipAsset.hpp	22
FemArchiver	Assets	Source/Engine/ Assets/ FemArchiver.hpp	131
FempakInput	Assets	Source/Engine/ Assets/ FempakWriter.hpp	15
FempakWriter	Assets	Source/Engine/ Assets/ FempakWriter.hpp	22
LocaleTable	Assets	Source/Engine/ Assets/ Localization.hpp	14
LocalizationGender	Assets	Source/Engine/	21

		Assets/ Localization.hpp	
LocalizationPack	Assets	Source/Engine/ Assets/ Localization.hpp	13
LocalizationTable	Assets	Source/Engine/ Assets/ Localization.hpp	28
MappedFile	Assets	Source/Engine/ Assets/VFS.hpp	20
MountRecord	Assets	Source/Engine/ Assets/VFS.hpp	134
PakEntry	Assets	Source/Engine/ Assets/ PakFormat.hpp	26
PakHeader	Assets	Source/Engine/ Assets/ PakFormat.hpp	16
PcmFormat	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	21
PinnedFile	Assets	Source/Engine/ Assets/VFS.hpp	67
ReloadNotice	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	94
SpriteAtlas	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	47
SpriteAtlasAsset	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	77
StagedRow	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	116
StringEntry	Assets	Source/Engine/ Assets/ Localization.hpp	15
TextureAlphaMode	Assets	Source/Engine/ Assets/ TextureFormat.hpp	43
TextureArrayAsset	Assets	Source/Engine/ Assets/ TextureArray.hpp	51
TextureCookOptions	Assets	Source/Engine/ Assets/	31

		TextureCook.hpp	
TextureFormat	Assets	Source/Engine/ Assets/ TextureFormat.hpp	22
TextureHeader	Assets	Source/Engine/ Assets/ TextureFormat.hpp	184
VfsEntry	Assets	Source/Engine/ Assets/VFS.hpp	59
VirtualFileSystem	Assets	Source/Engine/ Assets/ TextureArray.hpp	16
VirtualFileSystem	Assets	Source/Engine/ Assets/VFS.hpp	73

Engine/Audio (122)

Symbol	Kind	Header	Line
Applied	Audio	Source/Engine/ Audio/ VoicePreset.hpp	154
AudioBus	Audio	Source/Engine/ Audio/Audio.hpp	27
AudioBusComponent	Audio	Source/Engine/ Audio/Audio.hpp	44
AudioCaptureMonitor	Audio	Source/Engine/ Audio/ AudioDevices.hpp	144
AudioClip	Audio	Source/Engine/ Audio/Audio.hpp	75
AudioCommand	Audio	Source/Engine/ Audio/DspGraph.hpp	138
AudioCommandType	Audio	Source/Engine/ Audio/DspGraph.hpp	121
AudioContextComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	69
AudioDeviceEnumerator	Audio	Source/Engine/ Audio/ AudioDevices.hpp	93
AudioDeviceInfo	Audio	Source/Engine/ Audio/ AudioDevices.hpp	54

AudioDeviceKind	Audio	Source/Engine/ Audio/ AudioDevices.hpp	15
AudioDeviceList	Audio	Source/Engine/ Audio/ AudioDevices.hpp	66
AudioDevicePreferences	Audio	Source/Engine/ Audio/ AudioDevices.hpp	73
AudioDeviceStatus	Audio	Source/Engine/ Audio/ AudioDevices.hpp	21
AudioLimiter	Audio	Source/Engine/ Audio/Limiter.hpp	23
AudioListener	Audio	Source/Engine/ Audio/Audio.hpp	100
AudioSource	Audio	Source/Engine/ Audio/Audio.hpp	85
AudioSourceComponent	Audio	Source/Engine/ Audio/Audio.hpp	57
AudioSystem	Audio	Source/Engine/ Audio/Audio.hpp	106
AudioSystem	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	15
AudioZoneComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	34
AudioZoneKind	Audio	Source/Engine/ Audio/ AudioZones.hpp	14
AudioZoneSystem	Audio	Source/Engine/ Audio/ AudioZones.hpp	93
BiquadState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	369
BusAtomics	Audio	Source/Engine/ Audio/DspGraph.hpp	370
BusFilterParams	Audio	Source/Engine/ Audio/DspGraph.hpp	108
BusRuntime	Audio	Source/Engine/ Audio/Audio.hpp	227
BusShadow	Audio	Source/Engine/	291

		Audio/DspGraph.hpp	
BusState	Audio	Source/Engine/ Audio/DspGraph.hpp	338
CommandQueue	Audio	Source/Engine/ Audio/DspGraph.hpp	170
CookedClipRecord	Audio	Source/Engine/ Audio/Audio.hpp	202
DspClipView	Audio	Source/Engine/ Audio/DspGraph.hpp	205
DspGraph	Audio	Source/Engine/ Audio/DspGraph.hpp	216
DspGraph	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	16
DspResult	Audio	Source/Engine/ Audio/DspGraph.hpp	34
DspResult	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	19
DuckingParams	Audio	Source/Engine/ Audio/DspGraph.hpp	80
DuckingState	Audio	Source/Engine/ Audio/DspGraph.hpp	298
Entry	Audio	Source/Engine/ Audio/ VoicePreset.hpp	83
FilterState	Audio	Source/Engine/ Audio/DspGraph.hpp	327
FloatRing	Audio	Source/Engine/ Audio/Audio.hpp	25
FloatRing	Audio	Source/Engine/ Audio/ AudioStream.hpp	16
FloatRing	Audio	Source/Engine/ Audio/DspGraph.hpp	32
JitterSlot	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	119
LimiterDesc	Audio	Source/Engine/ Audio/Limiter.hpp	11
OggOpusStream	Audio	Source/Engine/ Audio/OggOpus.hpp	131

OggPacket	Audio	Source/Engine/ Audio/OggOpus.hpp	76
OggPacketReader	Audio	Source/Engine/ Audio/OggOpus.hpp	86
OggPage	Audio	Source/Engine/ Audio/OggOpus.hpp	36
OggPageReader	Audio	Source/Engine/ Audio/OggOpus.hpp	57
OpusHeadInfo	Audio	Source/Engine/ Audio/OggOpus.hpp	106
OwnedPcmRecord	Audio	Source/Engine/ Audio/Audio.hpp	209
Peer	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	131
PlayerState	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	88
RateBucket	Audio	Source/Engine/ Audio/ VoiceNetworkBridge.hpp	62
ResolvedZone	Audio	Source/Engine/ Audio/ AudioZones.hpp	115
ReverbParams	Audio	Source/Engine/ Audio/DspGraph.hpp	96
ReverbState	Audio	Source/Engine/ Audio/DspGraph.hpp	311
SlotRuntime	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	385
SourceRuntime	Audio	Source/Engine/ Audio/Audio.hpp	219
State	Audio	Source/Engine/ Audio/ AudioDevices.hpp	164
Stats	Audio	Source/Engine/ Audio/ AudioZones.hpp	96
VoiceAec	Audio	Source/Engine/ Audio/VoiceAec.hpp	44
VoiceAecSettings	Audio	Source/Engine/	22

		Audio/VoiceAec.hpp	
VoiceCapture	Audio	Source/Engine/ Audio/ AudioStream.hpp	64
VoiceChannel	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	65
VoiceChannelConfig	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	36
VoiceChannelStats	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	46
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GameplayEffectSystem	ECS	Source/Engine/ECS/SystemContext.hpp	24
GameplayServices	ECS	Source/Engine/ECS/SystemContext.hpp	107
GameplaySkillSystem	ECS	Source/Engine/ECS/SystemContext.hpp	25
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GameplaySystemView	ECS	Source/Engine/ECS/SystemContext.hpp	284
GameplayTagRegistry	ECS	Source/Engine/ECS/SystemContext.hpp	22
GpuEntity	ECS	Source/Engine/ECS/RenderEcs.hpp	31
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Light2DTargetComponent	ECS	Source/Engine/ECS/Lighting.hpp	53
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LightGridInput	ECS	Source/Engine/ECS/LightGrid.hpp	16
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LightingSystem	ECS	Source/Engine/ECS/Lighting.hpp	147
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PagedNavGrid	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	32
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NetInputStatus	Network	Source/Engine/ Network/ NetRollback.hpp	99
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NetTickMode	Network	Source/Engine/ Network/Session.hpp	40
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RollbackReport	Network	Source/Engine/ Network/ NetRollback.hpp	229
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SecureChannel	Network	Source/Engine/ Network/ SecureChannel.hpp	198
SecureChannelConfig	Network	Source/Engine/ Network/ SecureChannel.hpp	151
SecureCookieIssuer	Network	Source/Engine/ Network/ SecureChannel.hpp	131
SecureEndpoint	Network	Source/Engine/ Network/ SecureEndpoint.hpp	16
SecureSessionAuthConfig	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	24
SecureSessionTransport	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	38
SecureStatus	Network	Source/Engine/ Network/ SecureChannel.hpp	65
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SnapshotReceiver	Network	Source/Engine/ Network/ Replication.hpp	177
SnapshotReceiverConfig	Network	Source/Engine/ Network/ Replication.hpp	166
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ChainError	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	146
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ClothFlags	Physics	Source/Engine/ Physics/Cloth.hpp	20
ClothHandle	Physics	Source/Engine/ Physics/Cloth.hpp	46
ClothInfo	Physics	Source/Engine/ Physics/Cloth.hpp	99
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FluidDesc	Physics	Source/Engine/ Physics/Fluid.hpp	114
FluidInteractorComponent	Physics	Source/Engine/ Physics/Fluid.hpp	210
FluidWaveSystem	Physics	Source/Engine/ Physics/Fluid.hpp	221
GerstnerField	Physics	Source/Engine/ Physics/Fluid.hpp	49
GerstnerSample	Physics	Source/Engine/ Physics/Fluid.hpp	39
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GravityBodyShape	Physics	Source/Engine/Physics/Gravity.hpp	41
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GravitySystem	Physics	Source/Engine/Physics/Gravity.hpp	124
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GroupIndexEntry	Physics	Source/Engine/Physics/TransportGroups.hpp	127
JointBreakReason	Physics	Source/Engine/Physics/PhysicsJoints.hpp	72
JointChainComponent	Physics	Source/Engine/Physics/PhysicsJoints.hpp	378
JointChainDesc	Physics	Source/Engine/Physics/PhysicsJoints.hpp	187
JointChainFlags	Physics	Source/Engine/Physics/PhysicsJoints.hpp	91
JointChainHandle	Physics	Source/Engine/Physics/PhysicsJoints.hpp	250
JointChainInfo	Physics	Source/Engine/Physics/PhysicsJoints.hpp	429
JointChainKind	Physics	Source/Engine/Physics/PhysicsJoints.hpp	54
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JointConnectionInfo	Physics	Source/Engine/Physics/PhysicsJoints.hpp	287
JointConnectionKind	Physics	Source/Engine/Physics/PhysicsJoints.hpp	64
JointConnectionParams	Physics	Source/Engine/Physics/PhysicsJoints.hpp	559
JointLinkComponent	Physics	Source/Engine/Physics/PhysicsJoints.hpp	33
JointLinkFlags	Physics	Source/Engine/Physics/PhysicsJoints.hpp	22
JointTieDesc	Physics	Source/Engine/Physics/PhysicsJoints.hpp	328
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LevelGravityComponent	Physics	Source/Engine/Physics/Gravity.hpp	55
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ParticleEmitterFlags	Physics	Source/Engine/Physics/Particles.hpp	18
ParticleEmitterGpu	Physics	Source/Engine/Physics/Particles.hpp	69
ParticleSeedPush	Physics	Source/Engine/Physics/Particles.hpp	92
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PhysicsDesc	Physics	Source/Engine/Physics/Physics.hpp	154
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PhysicsSystem	Physics	Source/Engine/Physics/TransportGroups.hpp	13
Projectile2D	Physics	Source/Engine/Physics/Physics.hpp	143
ProjectileHitEvent	Physics	Source/Engine/Physics/ProjectileSystem.hpp	12
ProjectileSystem	Physics	Source/Engine/Physics/ProjectileSystem.hpp	24
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SdfBrickMap	Physics	Source/Engine/ Physics/ SdfCollision.hpp	106
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SdfBrickState	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	24
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SdfCarveReplayContext	Physics	Source/Engine/ Physics/ SdfDestructionCommands.hpp	37
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SdfCharacterComponent	Physics	Source/Engine/ Physics/ SdfCharacter.hpp	41
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SdfCharacterMeshData	Physics	Source/Engine/ Physics/ SdfCharacter.hpp	110
SdfCollisionDesc	Physics	Source/Engine/ Physics/ SdfCollision.hpp	49
SdfCollisionSystem	Physics	Source/Engine/ Physics/ SdfCollision.hpp	109

SdfComposeLayer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	25
SdfComposeOp	Physics	Source/Engine/ Physics/ SdfComposer.hpp	16
SdfComposer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	37
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BookSpan	Text	Source/Engine/Text/ BookAsset.hpp	77
BookStyle	Text	Source/Engine/Text/ BookAsset.hpp	86
BookViewport	Text	Source/Engine/Text/ BookSession.hpp	19
ChainedShaper	Text	Source/Engine/Text/ TextShaper.hpp	133
CodepointShaper	Text	Source/Engine/Text/ TextShaper.hpp	99
Entry	Text	Source/Engine/Text/ TextShaper.hpp	247
FontCandidate	Text	Source/Engine/Text/ TextFontStack.hpp	24
FontSlot	Text	Source/Engine/Text/ TextFontStack.hpp	59
FormatEntry	Text	Source/Engine/Text/ TextService.hpp	583
FormattedLayoutEntry	Text	Source/Engine/Text/ TextService.hpp	566
FormattedLayoutKey	Text	Source/Engine/Text/ TextService.hpp	556
HarfBuzzShaper	Text	Source/Engine/Text/ TextShaper.hpp	110
Impl	Text	Source/Engine/Text/ TextShaper.hpp	128
ITextShaper	Text	Source/Engine/Text/ TextShaper.hpp	89
LayoutEntry	Text	Source/Engine/Text/ TextService.hpp	548
LayoutKey	Text	Source/Engine/Text/ TextService.hpp	528
MappedTextAsset	Text	Source/Engine/Text/ TextAssetFile.hpp	17

MarkupStackEntry	Text	Source/Engine/Text/TextMarkup.hpp	71
MarkupVariantEntry	Text	Source/Engine/Text/TextService.hpp	574
ScriptBackend	Text	Source/Engine/Text/TextShapeSystem.hpp	124
ScriptShaper	Text	Source/Engine/Text/TextShaper.hpp	207
ShapedGlyph	Text	Source/Engine/Text/TextShaper.hpp	56
ShapedGlyphHit	Text	Source/Engine/Text/TextShapedLayout.hpp	153
ShapedLayoutResult	Text	Source/Engine/Text/TextShapedLayout.hpp	26
ShapedLine	Text	Source/Engine/Text/TextShapedLayout.hpp	18
StyledAsset	Text	Source/Engine/Text/TextService.hpp	595
TextAction	Text	Source/Engine/Text/TextStyle.hpp	61
TextActionKind	Text	Source/Engine/Text/TextStyle.hpp	51
TextAnimationParams	Text	Source/Engine/Text/TextRender.hpp	29
TextAssetBlock	Text	Source/Engine/Text/TextAssetFormat.hpp	34
TextAssetHeader	Text	Source/Engine/Text/TextAssetFormat.hpp	20
TextAssetRef	Text	Source/Engine/Text/TextAsset.hpp	14
TextAssetStatus	Text	Source/Engine/Text/TextAssetFormat.hpp	63
TextAssetView	Text	Source/Engine/Text/TextAssetFormat.hpp	42
TextBlock	Text	Source/Engine/Text/TextAsset.hpp	37
TextDirection	Text	Source/Engine/Text/TextShaper.hpp	21

TextFontStack	Text	Source/Engine/Text/TextFontStack.hpp	54
TextFormatArg	Text	Source/Engine/Text/TextFormat.hpp	45
TextFormatStatus	Text	Source/Engine/Text/TextFormat.hpp	33
TextGradientParams	Text	Source/Engine/Text/TextRender.hpp	22
TextInlineHit	Text	Source/Engine/Text/TextLayout.hpp	204
TextInlineKind	Text	Source/Engine/Text/TextStyle.hpp	73
TextInlineObject	Text	Source/Engine/Text/TextStyle.hpp	83
TextInlineSprite	Text	Source/Engine/Text/TextStyle.hpp	99
TextLayoutParams	Text	Source/Engine/Text/TextLayout.hpp	54
TextLayoutResult	Text	Source/Engine/Text/TextLayout.hpp	72
TextLine	Text	Source/Engine/Text/TextLayout.hpp	22
TextMarkupStatus	Text	Source/Engine/Text/TextMarkup.hpp	16
TextRenderEffects	Text	Source/Engine/Text/TextRender.hpp	44
TextRenderParams	Text	Source/Engine/Text/TextRender.hpp	57
TextRenderStats	Text	Source/Engine/Text/TextRender.hpp	76
TextRenderSystemStats	Text	Source/Engine/Text/TextRenderSystem.hpp	19
TextRun	Text	Source/Engine/Text/TextShaper.hpp	47
TextRuntime	Text	Source/Engine/Text/TextRuntime.hpp	17
TextScript	Text	Source/Engine/Text/TextShaper.hpp	30
TextService	Text	Source/Engine/Text/TextService.hpp	18
TextShapeCache	Text	Source/Engine/Text/	59

		TextShapeCache.hpp	
TextShapeHandle	Text	Source/Engine/Text/TextShapeCache.hpp	19
TextShapeKey	Text	Source/Engine/Text/TextShapeCache.hpp	32
TextShapeParams	Text	Source/Engine/Text/TextShaper.hpp	71
TextShapeResult	Text	Source/Engine/Text/TextShaper.hpp	82
TextShapeSystemStats	Text	Source/Engine/Text/TextShapeSystem.hpp	18
TextShapeView	Text	Source/Engine/Text/TextShapeCache.hpp	50
TextSpan	Text	Source/Engine/Text/TextStyle.hpp	40
TextStore	Text	Source/Engine/Text/TextStore.hpp	19
TextStyle	Text	Source/Engine/Text/TextStyle.hpp	27
TextStyledBlock	Text	Source/Engine/Text/TextMarkup.hpp	30
TextStyleFlags	Text	Source/Engine/Text/TextStyle.hpp	12
UnicodeBoundary	Text	Source/Engine/Text/UnicodeBreak.hpp	13
VerticalFormEntry	Text	Source/Engine/Text/TextVertical.hpp	16

Engine/UI (115)

Symbol	Kind	Header	Line
AccessibilityEditorSettings	UI	Source/Engine/UI/Accessibility.hpp	251
AccessibilityEffectKind	UI	Source/Engine/UI/AccessibilityEffects.hpp	26
AccessibilityEffectProfile	UI	Source/Engine/UI/AccessibilityEffects.hpp	84
AccessibilityEffectRegistry	UI	Source/Engine/UI/AccessibilityEffects.hp	122

		p	
AccessibilityEffectTag	UI	Source/Engine/UI/AccessibilityEffects.hpp	58
AccessibilityExtensionRegistry	UI	Source/Engine/UI/AccessibilityEffects.hpp	103
AccessibilityFlags	UI	Source/Engine/UI/Accessibility.hpp	209
AccessibilityKernel	UI	Source/Engine/UI/Accessibility.hpp	305
AccessibilityOverlay	UI	Source/Engine/UI/AccessibilityOverlay.hpp	30
AccessibilityPaletteExtension	UI	Source/Engine/UI/AccessibilityEffects.hpp	42
AccessibilityPresentationSettings	UI	Source/Engine/UI/Accessibility.hpp	234
AccessibilityProfile	UI	Source/Engine/UI/Accessibility.hpp	273
AccessibilitySettings	UI	Source/Engine/UI/Accessibility.hpp	218
AccessibilitySlot	UI	Source/Engine/UI/AccessibilityTheme.hpp	61
AccessibilityTemperatureBand	UI	Source/Engine/UI/AccessibilityEffects.hpp	49
AccessibilityTextStyle	UI	Source/Engine/UI/AccessibilityTheme.hpp	82
AccessibilityTheme	UI	Source/Engine/UI/AccessibilityTheme.hpp	92
AccessibilityThemeRegistry	UI	Source/Engine/UI/AccessibilityTheme.hpp	116
AlcoholEffect	UI	Source/Engine/UI/AccessibilityEffects.hpp	63
AlcoholEffectLayer	UI	Source/Engine/UI/AccessibilityEffects.hpp	77

AudioSystem	UI	Source/Engine/UI/Accessibility.hpp	27
AvSpeechTts	UI	Source/Engine/UI/Tts.hpp	60
BuildStatus	UI	Source/Engine/UI/PinkUI.hpp	1446
CardBoardComponent	UI	Source/Engine/UI/CardBoard.hpp	49
CardBoardSystem	UI	Source/Engine/UI/CardBoard.hpp	74
ColorMatrix3	UI	Source/Engine/UI/Accessibility.hpp	43
ColorVisionDeficiency	UI	Source/Engine/UI/Accessibility.hpp	35
Dropdown	UI	Source/Engine/UI/PinkUI.hpp	535
FlashGuard	UI	Source/Engine/UI/Accessibility.hpp	60
FocusDirection	UI	Source/Engine/UI/PinkUI.hpp	212
FocusPolicy	UI	Source/Engine/UI/PinkUI.hpp	188
FontAtlas	UI	Source/Engine/UI/FontAtlas.hpp	101
FontAtlasHeader	UI	Source/Engine/UI/FontAtlas.hpp	45
FontAtlasSet	UI	Source/Engine/UI/FontAtlas.hpp	167
FontBuildOptions	UI	Source/Engine/UI/FontAtlas.hpp	25
FontBuildOptions	UI	Source/Engine/UI/FontBuilder.hpp	15
FontCoverage	UI	Source/Engine/UI/FontCoverage.hpp	21
FontFamilyFile	UI	Source/Engine/UI/FontAtlas.hpp	74
FontFamilyManifest	UI	Source/Engine/UI/FontFamilyAsset.hpp	17
FontPixelFormat	UI	Source/Engine/UI/FontAtlas.hpp	37
FontStyle	UI	Source/Engine/UI/	65

		FontAtlas.hpp	
FrameOutput	UI	Source/Engine/UI/PinkUI.hpp	1421
GlyphRecord	UI	Source/Engine/UI/FontAtlas.hpp	80
GridView	UI	Source/Engine/UI/PinkUI.hpp	425
HyperlinkRange	UI	Source/Engine/UI/Hypertext.hpp	51
HypertextComponent	UI	Source/Engine/UI/Hypertext.hpp	32
HypertextFlags	UI	Source/Engine/UI/Hypertext.hpp	21
HypertextPool	UI	Source/Engine/UI/Hypertext.hpp	74
HypertextView	UI	Source/Engine/UI/Hypertext.hpp	67
ITextShaper	UI	Source/Engine/UI/PinkUI.hpp	34
LayoutCache	UI	Source/Engine/UI/PinkUI.hpp	741
LazyFontFamily	UI	Source/Engine/UI/FontFamilyAsset.hpp	101
ListView	UI	Source/Engine/UI/PinkUI.hpp	561
LocalizationTable	UI	Source/Engine/UI/Accessibility.hpp	25
LocalizationTable	UI	Source/Engine/UI/Hypertext.hpp	16
LocalizationTable	UI	Source/Engine/UI/PinkUI.hpp	40
NarrativeArchetype	UI	Source/Engine/UI/NarrativeTree.hpp	23
NarrativeEvent	UI	Source/Engine/UI/Accessibility.hpp	84
NarrativeGroup	UI	Source/Engine/UI/NarrativeTree.hpp	37
NarrativeLocale	UI	Source/Engine/UI/NarrativeTree.hpp	17
NarrativeLog	UI	Source/Engine/UI/Accessibility.hpp	100

NarrativeNode	UI	Source/Engine/UI/ NarrativeTree.hpp	47
NarrativeTree	UI	Source/Engine/UI/ NarrativeTree.hpp	55
NvgPattern	UI	Source/Engine/UI/ AccessibilityEffects.hp p	33
OledShiftState	UI	Source/Engine/UI/ PinkUI.hpp	464
PinkSdfDraw	UI	Source/Engine/UI/ PinkSdf.hpp	74
PinkSdfFlags	UI	Source/Engine/UI/ PinkSdf.hpp	24
PinkSdfGpuPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	51
PinkSdfPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	33
PinkSdfShape	UI	Source/Engine/UI/ PinkSdf.hpp	15
RadialProgress	UI	Source/Engine/UI/ PinkUI.hpp	585
RetainedFont	UI	Source/Engine/UI/ FontAtlas.hpp	293
SapiTts	UI	Source/Engine/UI/ Tts.hpp	30
ScrollView	UI	Source/Engine/UI/ PinkUI.hpp	492
SlotComponent	UI	Source/Engine/UI/ CardBoard.hpp	27
SlotFlags	UI	Source/Engine/UI/ CardBoard.hpp	15
SlotLayout	UI	Source/Engine/UI/ CardBoard.hpp	65
SoundCueCategory	UI	Source/Engine/UI/ SoundCues.hpp	26
SoundCueDef	UI	Source/Engine/UI/ SoundCues.hpp	62
SoundCueRegistry	UI	Source/Engine/UI/ SoundCues.hpp	71
SoundTrigger	UI	Source/Engine/UI/ Accessibility.hpp	129
StateTrack	UI	Source/Engine/UI/	424

		Accessibility.hpp	
SubtitleLine	UI	Source/Engine/UI/ Accessibility.hpp	140
SubtitleQueue	UI	Source/Engine/UI/ Accessibility.hpp	164
SystemContext	UI	Source/Engine/UI/ PinkUI.hpp	29
TabBar	UI	Source/Engine/UI/ PinkUI.hpp	516
TextAlign	UI	Source/Engine/UI/ PinkUI.hpp	223
TextShapeParams	UI	Source/Engine/UI/ PinkUI.hpp	35
TextVerticalAlign	UI	Source/Engine/UI/ PinkUI.hpp	230
TimedSoundCue	UI	Source/Engine/UI/ Accessibility.hpp	153
TooltipPlacer	UI	Source/Engine/UI/ PinkUI.hpp	596
TtsBackend	UI	Source/Engine/UI/ Accessibility.hpp	28
TtsBackend	UI	Source/Engine/UI/ Tts.hpp	19
TtsPriority	UI	Source/Engine/UI/ Tts.hpp	11
UiPalette	UI	Source/Engine/UI/ UiTheme.hpp	41
UiPaletteSlot	UI	Source/Engine/UI/ UiTheme.hpp	18
UiThemeAsset	UI	Source/Engine/UI/ UiTheme.hpp	46
VirtualFileSystem	UI	Source/Engine/UI/ FontAtlas.hpp	19
VoiceCaptionEvent	UI	Source/Engine/UI/ Accessibility.hpp	194
VoiceCaptionProvider	UI	Source/Engine/UI/ Accessibility.hpp	202
WidgetCanvas	UI	Source/Engine/UI/ Accessibility.hpp	31
WidgetCanvas	UI	Source/Engine/UI/ PinkUI.hpp	291

WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	484
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	608
WidgetCanvasSystem	UI	Source/Engine/UI/PinkUI.hpp	1459
WidgetComponent	UI	Source/Engine/UI/PinkUI.hpp	252
WidgetComponentFlags	UI	Source/Engine/UI/PinkUI.hpp	264
WidgetDesc	UI	Source/Engine/UI/PinkUI.hpp	310
WidgetFlags	UI	Source/Engine/UI/PinkUI.hpp	146
WidgetHandle	UI	Source/Engine/UI/PinkUI.hpp	81
WidgetHyperlink	UI	Source/Engine/UI/PinkUI.hpp	301
WidgetKind	UI	Source/Engine/UI/PinkUI.hpp	100
WidgetLayoutAsset	UI	Source/Engine/UI/WidgetLayoutAsset.hpp	15
WidgetPaintContext	UI	Source/Engine/UI/PinkUI.hpp	292
WidgetPaintContext	UI	Source/Engine/UI/PinkUI.hpp	392

Engine/Video (28)

Symbol	Kind	Header	Line
AssetDatabase	Video	Source/Engine/Video/VideoStream.hpp	28
AudioPumpStatus	Video	Source/Engine/Video/VideoAudioStream.hpp	15
ClusterIndexEntry	Video	Source/Engine/Video/WebmDemuxer.hpp	210
ClusterScan	Video	Source/Engine/Video/	150

		WebmDemuxer.hpp	
ContainerKind	Video	Source/Engine/ Video/ WebmDemuxer.hpp	225
DecodeStatus	Video	Source/Engine/ Video/VideoTypes.hpp	33
DemuxStatus	Video	Source/Engine/ Video/ WebmDemuxer.hpp	74
Element	Video	Source/Engine/ Video/ WebmDemuxer.hpp	127
FrameBuffer	Video	Source/Engine/ Video/ VideoPlayer.hpp	160
FrameSlot	Video	Source/Engine/ Video/ VideoPlayer.hpp	167
LacedAudioPacket	Video	Source/Engine/ Video/ WebmDemuxer.hpp	201
MappedFile	Video	Source/Engine/ Video/ VideoPlayer.hpp	24
RHI	Video	Source/Engine/ Video/ VideoPlayer.hpp	29
SystemGraph	Video	Source/Engine/ Video/ VideoStream.hpp	24
VideoAudioDecoder	Video	Source/Engine/ Video/ VideoAudioDecoder.hpp	15
VideoAudioStream	Video	Source/Engine/ Video/ VideoAudioStream.hpp	27
VideoPlayer	Video	Source/Engine/ Video/ VideoPlayer.hpp	35
VideoStreamComponent	Video	Source/Engine/ Video/ VideoStream.hpp	36
VideoStreamFlags	Video	Source/Engine/	20

		Video/VideoTypes.hpp	
VideoSystem	Video	Source/Engine/ Video/ VideoStream.hpp	56
VirtualFileSystem	Video	Source/Engine/ Video/ VideoStream.hpp	20
Vp9Backend	Video	Source/Engine/ Video/ Vp9Backend.hpp	30
WebmAudioPacket	Video	Source/Engine/ Video/ WebmDemuxer.hpp	68
WebmAudioTrackInfo	Video	Source/Engine/ Video/ WebmDemuxer.hpp	43
WebmDemuxer	Video	Source/Engine/ Video/ WebmDemuxer.hpp	82
WebmFrame	Video	Source/Engine/ Video/ WebmDemuxer.hpp	54
WebmTrackInfo	Video	Source/Engine/ Video/ WebmDemuxer.hpp	15
YuvFrameView	Video	Source/Engine/ Video/ Vp9Backend.hpp	12

Engine/Weather (12)

Symbol	Kind	Header	Line
ClothSystem	Weather	Source/Engine/ Weather/Weather.hpp	13
WeatherAtmosphere Pass	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	128
WeatherContextCom ponent	Weather	Source/Engine/ Weather/Weather.hpp	32
WeatherDesc	Weather	Source/Engine/ Weather/Weather.hpp	91
WeatherFlags	Weather	Source/Engine/ Weather/Weather.hpp	19

WeatherGpuConstants	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	46
WeatherGpuResources	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	18
WeatherGpuStats	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	68
WeatherPreconciliatedDesc	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	32
WeatherPreconciliatedSystem	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	78
WeatherSystem	Weather	Source/Engine/Weather/Weather.hpp	99
WeatherWindField	Weather	Source/Engine/Weather/Weather.hpp	75

21. Appendix B - macros and platform defines

Macro	Header	Line
FBY_ALCOHOL_EFFECTS	Source/Engine/Core/PluginAPI.h	105
FBY_AOT_EXPORT	Source/Engine/Script/AotAbi.hpp	519
FBY_AOT_EXPORT	Source/Engine/Script/AotAbi.hpp	521
FBY_ASSERT	Source/Engine/Core/Assert.hpp	13
FBY_ASSERT	Source/Engine/Core/Assert.hpp	31
FBY_ASSERT_MSG	Source/Engine/Core/Assert.hpp	22
FBY_ASSERT_MSG	Source/Engine/Core/Assert.hpp	32
FBY_DEFINE_PLUGIN	Source/Engine/Core/	55

	Plugin.hpp	
FBY_DETAIL_CONCAT	Source/Engine/Core/ Definitions.hpp	263
FBY_DETAIL_CONCAT_INNER	Source/Engine/Core/ Definitions.hpp	262
FBY_EFFECT_TAGS	Source/Engine/Core/ PluginAPI.h	104
FBY_FIELD	Source/Engine/Core/ Mirror.hpp	288
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/ Fluid.hpp	16
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/ Fluid.hpp	18
FBY_HAS_MDSPAN	Source/Engine/ECS/ Skeleton.hpp	18
FBY_HAS_MDSPAN	Source/Engine/ECS/ Skeleton.hpp	20
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	105
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	108
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	112
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/ FlatMap.hpp	19
FBY_PALETTE_EXTENSION_COLORS	Source/Engine/Core/ PluginAPI.h	103
FBY_PLATFORM_ABI_VERSION	Source/Plugins/ PlatformService/PlatformAPI.h	11
FBY_PLATFORM_CALL	Source/Plugins/ PlatformService/PlatformAPI.h	15
FBY_PLATFORM_CALL	Source/Plugins/ PlatformService/PlatformAPI.h	18
FBY_PLATFORM_EXPORT	Source/Plugins/ PlatformService/PlatformAPI.h	14
FBY_PLATFORM_EXPORT	Source/Plugins/ PlatformService/PlatformAPI.h	17
FBY_PLUGIN_ABI_VERSION	Source/Engine/Core/ PluginAPI.h	30
FBY_PLUGIN_API_H	Source/Engine/Core/ PluginAPI.h	18

FBY_PLUGIN_COMPONENT	Source/Engine/Core/ Plugin.hpp	52
FBY_PLUGIN_ENTRY	Source/Engine/Core/ PluginAPI.h	214
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	33
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	35
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	37
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginApi.hpp	11
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginApi.hpp	13
FBY_PLUGIN_INFO_SYMBOL	Source/Engine/Core/ PluginAPI.h	206
FBY_PLUGIN_LOAD_SYMBOL	Source/Engine/Core/ PluginAPI.h	207
FBY_PLUGIN_UNLOAD_SYMBOL	Source/Engine/Core/ PluginAPI.h	208
FBY_PROJECT_ASSETS	Source/Engine/Assets/ ProjectScaffold.hpp	130
FBY_PROJECT_MAX_RENDER_ITEMS	Source/Engine/Assets/ ProjectScaffold.hpp	123
FBY_REFLECT_COMPONENT	Source/Engine/Core/ Mirror.hpp	295
FBY_REFLECT_COMPONENT_EMPTY	Source/Engine/Core/ Mirror.hpp	311
FBY_REGISTER_TYPE	Source/Engine/Core/ Definitions.hpp	266
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/ RelevanceGrid.hpp	19
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/ RelevanceGrid.hpp	21
FBY_RESTRICT	Source/Engine/Core/ Definitions.hpp	26
FBY_RESTRICT	Source/Engine/Core/ Definitions.hpp	28
FBY_SDF_SHAPE_FIELD	Source/Engine/Physics/ SdfPrimitives.hpp	172
FBY_TEMPERATURE_BAND	Source/Engine/Core/	106

S	PluginAPI.h	
FBY_THEME_COLOR_COUNT	Source/Engine/Core/PluginAPI.h	84
FBY_TYPE_NAME	Source/Engine/Core/Definitions.hpp	282
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	22
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	24
FBY_UNREACHABLE	Source/Engine/Core/Assert.hpp	45
FBY_VERIFY	Source/Engine/Core/Assert.hpp	36
FBY_VK_DECLARE	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	127
FBY_VK_DEVICE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	36
FBY_VK_GLOBAL_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	15
FBY_VK_INSTANCE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	20
FBY_VOICE_API_H	Source/Engine/Core/VoiceAPI.h	12
FBY_VOICE_API_VERSION	Source/Engine/Core/VoiceAPI.h	21
FBY_VOICE_FLAG_ALLOW_DOWNLINK	Source/Engine/Core/VoiceAPI.h	53
FBY_VOICE_FLAG_ALLOW_UPLINK	Source/Engine/Core/VoiceAPI.h	52
FBY_VOICE_FLAG_COSMETIC	Source/Engine/Core/VoiceAPI.h	50
FBY_VOICE_FLAG_MODERATION	Source/Engine/Core/VoiceAPI.h	51
FBY_VOICE_SLOT_COUNT	Source/Engine/Core/VoiceAPI.h	22
FORCEINLINE	Source/Engine/Core/Definitions.hpp	17
FORCEINLINE	Source/Engine/Core/Definitions.hpp	19
FORCEINLINE	Source/Engine/Core/Definitions.hpp	21